Complex Numbers: Roots of Equations

Questions 1–53

Question 1. A polynomial may have real roots or complex roots. For each of the following write down the possible combinations:

a) Cubic polynomial.
b) Quartic polynomial.
c) Quintic polynomial.

Question 2. Find all the roots of the equation

$$z^3 - 4z^2 + 30z - 52 = 0$$

given that one complex root is $1 + 5i$

Question 3. Find all the roots of the equation

$$-z^3 + 11z^2 - 64z + 170 = 0$$

given that one complex root is $3 - 5i$

Question 4. Given that

$$\mathrm{f}(z) = z^3 + az^2 + bz + c$$

where a, b and c are real numbers. It is given that $\mathrm{f}(z) = 0$ has solutions $z = 4$ and $z = 2 - 8i$
Find the values of a, b and c

Question 5. A cubic equation exists such that all its coefficients are real, state the equation given that two of its roots are -3 and $1+3i$

Question 6. Prerequisite knowledge for Q6: Argand Diagrams

A cubic f(z) has 3 distinct roots, which all lie on the same straight line in the complex plane. Given one root is $1+4i$

a) State the other two roots.

b) Find f(z)

Question 7. Given that

$$\mathrm{f}(z) = z^4 + az^3 + bz^2 + cz + d$$

where a, b, c and d are real numbers and $\mathrm{f}(z) = 0$ has roots $1 + i$, -2 and 3
Find the values of a, b, c and d

Question 8. A quartic equation $\mathrm{f}(z) = 0$ exists such that all its coefficients are real, find $\mathrm{f}(z)$ given that two of its roots are $1 - 5i$ and $2 + i$

Question 9. Given that

$$\mathrm{f}(z) = z^4 + az^3 + bz^2 + cz + d$$

where a, b, c and d are real numbers and that $\mathrm{f}(z) = 0$ has solutions $2 + i$ and $4 - 3i$
Find the values of a, b, c and d

Question 10. Find all the roots of the equation

$$z^3 - 13z^2 + 56z - 180 = 0$$

given that $z = 9$ is a solution.

Question 11. Find all the roots of the equation

$$z^3 - 10z^2 + 34z - 40$$

given that $(z - 4)$ is a factor.

Question 12. Given that $(z + 3)$ is a factor of $\mathrm{f}(z)$ where

$$\mathrm{f}(z) = z^3 + z^2 + 11z + 51$$

solve the equation $\mathrm{f}(z) = 0$

Question 13. Prerequisite knowledge for Q13-Q17: Argand Diagrams
Given that $z = 1 - 3i$ is a root of the equation

$$z^3 + 3z^2 + 50 = 0$$

Find the area of the triangle made by the three roots of the equation on an Argand diagram.

Question 14. Given that $z = 3$ is one solution of

$$z^3 - 7z^2 + 41z - 87 = 0$$

find the area of the triangle made by the three roots of the equation on an Argand diagram.

Question 15. Given that $z = 5 - \sqrt{3}i$ is a root of the equation

$$z^3 - 12z^2 + 48z - 56 = 0$$

a) find the area of the triangle made by the three roots of the equation on an Argand diagram.
b) Show that the triangle formed by the roots on the Argand diagram is equilateral.

Question 16. The roots of a quartic equation f(z) when plotted on an Argand diagram form a square of area 16 units^2. Each corner is in a different quadrant and one of the roots is $1 - 2i$

a) Sketch an Argand Diagram of all the solutions of f(z) = 0
b) Hence, determine f(z)

Question 17. The equation $f(z) = 0$, where $f(z)$ is a quartic polynomial with real coefficients, has a root $z = 3 + 5i$. Given that all the roots lie on a straight line in the complex plane and at least one of the roots is real:

a) Find the roots of $f(z)$

b) Hence, determine $f(z)$

Question 18. The function f is defined as

$$f(z) = z^3 - 7z^2 + 32z - 60$$

a) Find $(2 + 4i)^3$

b) Verify that $2 + 4i$ is a root of $f(z) = 0$

c) Express $f(z)$ as a product of a linear factor and a quadratic factor with real coefficients.

Question 19. The function f is defined by

$$\mathrm{f}(z) = z^4 - z^3 - 6z^2 + 14z - 12$$

a) Show that $z = 1 + i$ is a root of $\mathrm{f}(z) = 0$
b) Find the remaining roots.

Question 20. Given that $z = 2 - 7i$ is root of the equation

$$z^4 - 6z^3 + 71z^2 - 146z + 530 = 0$$

find all the roots of the equation.

Question 21. The function f is defined by

$$\mathrm{f}(z) = z^4 + 5z^3 + 6z^2 + 50z + 100$$

It is known that $z = 1 + 3i$ is a solution to $\mathrm{f}(z) = 0$

a) Find all the solutions to $\mathrm{f}(z) = 0$

b) Hence, given the curve $y = \mathrm{f}(z)$ only has one turning point, sketch $y = \mathrm{f}(z)$

Question 22. The function f is defined by

$$\mathrm{f}(z) = z^4 + 4z^3 + 3z^2 + 10z + 50$$

a) Show that $1 + 2i$ is a root of $\mathrm{f}(z) = 0$

b) Find all the roots of $\mathrm{f}(z) = 0$

c) Hence explain why the graph of $y = \mathrm{f}(x)$ does not intersect the x-axis.

Question 23. Find in exact form, showing all your steps, all the solutions of

a) $z^3 - 13z^2 + 56z - 78 = 0$

b) $z^3 - 6z^2 + 15z - 14 = 0$

c) $z^3 - 2z^2 - 2z + 12 = 0$

d) $3z^3 - 20z^2 + 54z - 28 = 0$

Question 24. Prerequisite knowledge for Q24-Q26: Argand Diagrams

A cubic polynomial f(z) with real coefficients has 3 distinct roots, which all lie on the same straight line in the complex plane. It is known that one root is $1 + 4i$

a) State the other two roots.

b) Find f(z)

Question 25. A quintic polynomial f(z) with real coefficients has 1 real root and 4 complex roots which all lie on the same straight line in the complex plane. It is known that two of the roots are $2+3i$ and $2-2i$

a) State all the roots.

b) Find f(z)

Question 26. Given that $2+i$ and $1-3i$ are roots of

$$z^4 + az^3 + bz^2 + cz + d = 0$$

where a, b, c and d are real constants.

a) Plot all the roots on an Argand diagram.

b) Find the area formed on the Argand diagram by the roots.

c) Find the values of a, b, c and d

Question 27. A complex number α is given to be $1 + 2i$
A quartic polynomial f(z) with real coefficients exists such that it has roots α and α^2

a) Find α^2

b) Find f(z)

Question 28. Given that

$$z^3 - 4z^2 + kz - 10 = 0$$

where k is a real constant, has a root $1 - 2i$

a) Find all the roots.

b) Find the value of k

Question 29. Given that

$$z^3 + kz^2 - 5z - 12 = 0$$

where k is a real constant, has a root $z = 4$

a) Find the value of k

b) Find all the roots.

Question 30. It is known that $3 - i$ is a root of

$$2z^3 - 19z^2 + pz + q = 0$$

where p and q are real constants.

a) Find the values of p and q

b) Find all the roots.

Question 31. It is known that $2 - i$ is a root of

$$z^4 - 6z^3 + pz^2 + qz + 20 = 0$$

where p and q are real integers.

a) Express

$$z^4 - 6z^3 + pz^2 + qz + 20$$

as a product of two quadratic factors.

b) Hence find the values of p and q

Question 32. Given that $2i$ is a root of the equation

$$z^4 - 3z^3 - 6z^2 + pz + q = 0$$

where p and q are real constants.

a) Find the values of p and q

b) Find the other roots of the equation.

Question 33. It is given that one of the roots of

$$z^4 - 6z^3 + 2z^2 - 54z - 63 = 0$$

is a purely imaginary.

a) Without first finding any other roots, find the purely imaginary roots.

b) Hence, find all the roots to the equation.

Question 34. It is given that

$$\mathrm{f}(z) = z^4 + az^3 + bz^2 + az + 13$$

where a and b are real numbers, and $\mathrm{f}(z) = 0$ has a solution $3 - 2i$. Express $\mathrm{f}(z)$ as a product of two quadratic factors with real coefficients.

Question 35. Given that $-1 + \frac{\sqrt{3}}{2}i$ is a root of the equation

$$4z^4 + 16z^3 + 51z^2 + kz + 49 = 0$$

where k is an integer,

a) find all the roots.

b) determine the value of k

Question 36. A complex number α is given to be $2 - i$

a) Find α^2

b) Find α^3

It is known that for real values p and q,

$$p\alpha^3 + 5\alpha^2 + 12\alpha + q = 0$$

c) Find the values of p and q

d) Hence, solve the equation

$$pz^3 + 5z^2 + 12z + q = 0$$

Question 37. Prerequisite knowledge for Q37: Modulus-Argument form and Argand Diagrams
Given that

$$f(z) = z^2 - 3z + 9$$

a) Find the roots of $f(z) = 0$ giving your answer in modulus-argument form
b) Given that the roots of $f(z) = 0$ are also roots of $z^3 + k = 0$ where k is a real integer, find the other root of $z^3 + k = 0$
c) Plot all the roots of $z^3 + k = 0$ on an Argand diagram.

Question 38. It is given that

$$z^3 + kz^2 - kz - 1 = 0$$

where k is real.
a) Find a real root of the equation.
b) Find the range of values of k such that the equation has a pair of distinct complex roots.

Question 39. The polynomial f(z) has real coefficients. One solution of f(z) = 0 is $5 + 2i$

a) Show that

$$z^2 - 10z + 29$$

is a factor of f(z)

b) Find all the solution to f(z) = 0 given that

$$\text{f}(z) = z^6 - 10z^5 + 29z^4 - 16z^2 + 160z - 464$$

Question 40. Prerequisite knowledge for Q40-Q46: Argand Diagrams

Given that $4 - 7i$ is a root of the equation

$$z^4 - 12z^3 + 117z^2 - 420z + 1300 = 0$$

Find the area of the quadrilateral made by the four roots of the equation on an Argand diagram.

Question 41. Find the area of the triangle when the three roots of the equation

$$z^3 - 12z^2 + 47z - 54 = 0$$

are plotted on an Argand Diagram.

Question 42. It is given that

$$z^4 + az^3 + bz^2 + cz + d = 0$$

where a, b, c and d are real numbers. When the roots of this equation are plotted on an Argand diagram they form a square with one vertex in each quadrant.
It is known that one root is $3 + 4i$

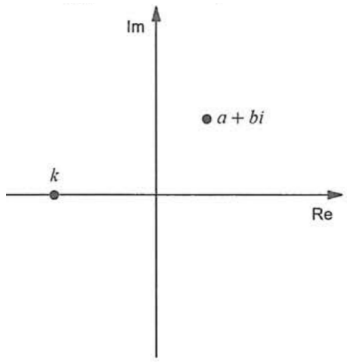
a) State the area of the square.
b) Determine the values of a, b, c and d

Question 43. For real numbers a and b the roots of

$$z^3 - 7z^2 + az + b = 0$$

form a triangle of area 56 when plotted on an Argand diagram.
Given that one of the roots is 3, find the values of a and b

Question 44. A cubic equation f(z) with real coefficients has two of its roots plotted on the Argand diagram below.
Find f(z) in terms of a, b and k



Question 45. Prerequisite knowledge for Q45: Complex Loci
Show that the roots of the equation

$$z^3 - 23z^2 + 299z - 2197 = 0$$

when plotted on an Argand diagram lie on the circle $|z| = k$, where k is a real number to be found.

Question 46. A cubic polynomial with real coefficients has 3 distinct roots. They are plotted on an Argand diagram and form an isosceles triangle with:

- an area of 42 units^2
- the distance between the complex roots which have real positive parts is 6 units
- the modulus of the complex roots is 5

Find the two possible cubic equations that satisfy these conditions.

Question 47. It is known that the equation

$$x^4 - x^3 + 8x^2 + 10x = 0$$

has a non-zero real root of a. Given that $a + 1 - 2i$ is a root of

$$z^4 + (a-2)z^2 + (13a - 15) = 0$$

Determine all the possible values of z

Question 48. The equation

$$z^4 - 8z^3 + 6z^2 + 148z - 272 = 0$$

has solutions $z = a$, $z = -b$ and $z = a + 3 + 3i$ where a and b are real positive integers.

a) State the fourth solution in terms of a

b) Find the value of a and the value of b

c) Hence state all the solutions to

$$x^4 - 8x^3 + 6x^2 + 148x - 272 = 0$$

Question 49. Prerequisite knowledge for Q49: Complex Loci
The function f is defined as

$$\mathrm{f}(z) = z^4 - 14z^3 + 74z^2 - 184z + 208$$

It is known that $\mathrm{f}(z) = 0$ has a root of $5 + i$

a) Solve $\mathrm{f}(z) = 0$

Two of the roots of $\mathrm{f}(z) = 0$ lie on the locus

$$C_1 = \{z : |z| = p\}$$

and the other two roots of $\mathrm{f}(z) = 0$ lie on the locus

$$C_2 = \{z : |z| = q\}$$

where $p > q$

b) Find the area of the region between C_1 and C_2 giving your answer in exact form.

Question 50. For each of the following, find all the roots and show that the sum of the roots of each of the following sum to zero:

a) $z^2 = 1$
b) $z^3 = 1$
c) $z^4 = 1$

Question 51. It is given that a cubic polynomial f(z) has real coefficients, and that the coefficient of the linear term is equal to the coefficient of the quadratic term. It has a complex root $p + qi$ and a real root k where p, q and k are real values.

a) Show that

$$k = -\frac{p^2 + q^2 + 2p}{1 + 2p}$$

b) Hence, given instead that f(z) has one real root and one repeated real root, show that

$$k = -\frac{p(p+2)}{2p+1}$$

Question 52. It is given that a cubic polynomial has real coefficients, and the linear term is zero.
Show that if it has a complex root $p + qi$ and a real root k where p, q and k are real values then $k^2 \geq q^2$

Question 53. It is given that a quartic polynomial has real coefficients, and the linear term is zero.
Show that if it has a complex root $p + qi$ and real roots m and n, where p, q, m and n are real non-zero constants, then

$$\frac{mn}{m+n} \geq q$$